



Introduction to Information and Communication Technologies CL1000

Week 04

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Recap

- Introduction to Number Systems
- Bit/Byte
- Number Conversions (binary, decimal, octal, hexadecimal)

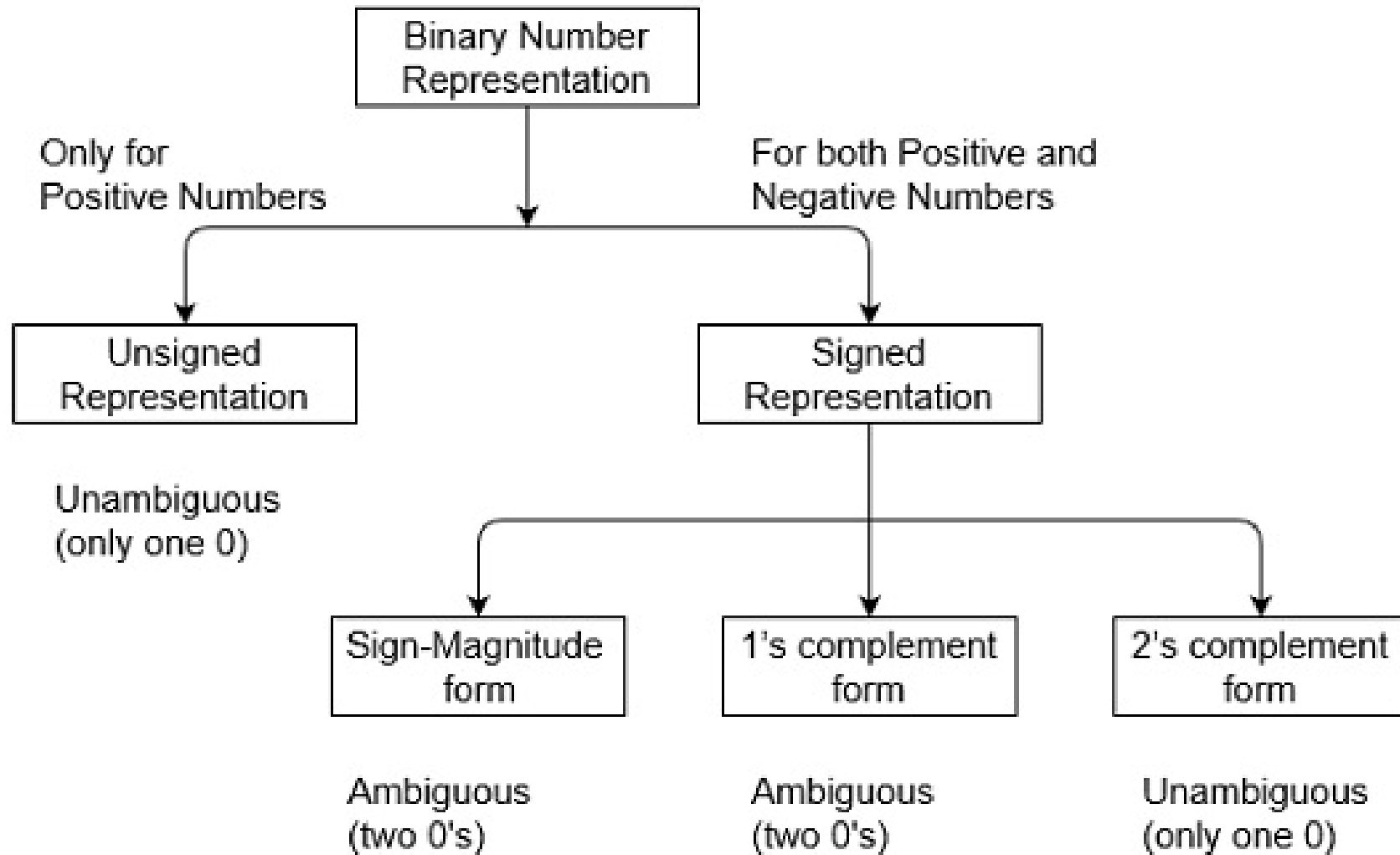
What Will You Learn Today?

- Integer Representations
- Fixed and Floating-point Representations
- Binary Arithmetic Operations (+, -, *, /)
- Complements in Number Systems (1's, 2's, 7's, 8's, 9's, 10's, 15's and 16's)

Integer Representation

Signed and Un-signed Binary Numbers

- Binary numbers can be represented in signed and unsigned way.
- Unsigned binary numbers do not have sign bit
- Signed binary numbers uses signed bit to represent positive and negative numbers



Unsigned numbers

- Don't have any sign, these can contain only magnitude of the number.
- Example:

$$(92)_{10} = (1 \times 2^6 + 0 \times 2^5 + 1 \times 2^4 + 1 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 0 \times 2^0)_{10}$$
$$= (1011100)_2$$

It's **7-bit binary magnitude** of the decimal number 92.

Range of Unsigned Numbers

- Range of unsigned binary number is from **0 to (2^n-1)**
- Range of 5-bit unsigned binary number: 0 to (2^5-1)

Signed numbers

- Contain sign flag, this representation distinguish positive and negative numbers.
- This technique contains both sign bit and magnitude of a number.

$$\begin{array}{ccccccc} 0 & 1 & 0 & 1 & 0 & 0 & 1 & = & +41 \\ \uparrow & & & & & & & & \\ \text{Sign bit} & & & & & & & & \end{array}$$
$$\begin{array}{ccccccc} 1 & 1 & 0 & 1 & 0 & 0 & 1 & = & -41 \\ \uparrow & & & & & & & & \\ \text{Sign bit} & & & & & & & & \end{array}$$

- There are several ways to represent signed binary numbers.
 1. Sign-Magnitude Form (the least used method)
 2. 1's complement
 3. 2's complement (the most used method)
- In all cases, the **MSB** in a signed number is the sign bit, that tells you if the number is positive or negative.

Sign-Magnitude Form

- Total Combinations 2^n
- Range of Sign-Magnitude form is from $-(2^{(n-1)}-1)$ to $(2^{(n-1)}-1)$

0 indicates +ve number

1 indicates -ve number

00111010 (true form).
Sign bit Magnitude bits

Decimal	Signed 1's compliment	Signed 2's compliment	Signed magnitude
7	111	111	111
6	110	110	110
5	101	101	101
4	100	100	100
3	11	11	11
2	10	10	10
1	1	1	1
0	0	0	0
0	1111	---	1000
-1	1110	1111	1001
-2	1101	1110	1010
-3	1100	1101	1011
-4	1011	1100	1100
-5	1010	1011	1101
-6	1001	1010	1110
-7	1001	1000	1111

1's complement

- The 1's complement of a binary number is obtained by changing each 0 to 1 and each 1 to 0.
- Range of 1's complement form is from $-(2^{(n-1)}-1)$ to $(2^{(n-1)}-1)$
- Find $(1100)_2$ 1's complement
Given 1 1 0 0
1's complement is 0 0 1 1
Result is $(0 0 1 1)_2$

2's Complement

- The 2's complement of a binary number is a binary number which is obtained by adding 1 to the 1's complement of a number i.e.

$$\text{2's complement} = \text{1's complement} + 1$$

- Range of 2's complement form is *from* $-(2^{(n-1)})$ to $(2^{(n-1)}-1)$
- Find $(1010)_2$ 2's complement

Given	1	0	1	0
1's complement is	0	1	0	1
				1
2's complement	0	1	1	0
Result is $(0110)_2$				

<i>decimal</i>	<i>binary</i>	<i>decimal</i>	<i>binary</i>
0	0000	0	0000000000000000
1	0001	1	0000000000000001
2	0010	2	0000000000000010
3	0011	3	0000000000000011
4	0100	...	...
5	0101	32765	0111111111111101
6	0110	32766	0111111111111110
7	0111	32767	0111111111111111
-8	1000	-32768	1000000000000000
-7	1001	-32767	1000000000000001
-6	1010	-32766	1000000000000010
-5	1011	-32765	1000000000000011
-4	1100	...	...
-3	1101	-3	1111111111111101
-2	1110	-2	1111111111111110
-1	1111	-1	1111111111111111
<i>4-bit integers</i> <i>(two's complement)</i>		<i>16-bit integers</i> <i>(two's complement)</i>	

Convert Floating-Point into Binary Number

- 4.25

Where,

4 is an integral part.

0.25 is a fractional part.

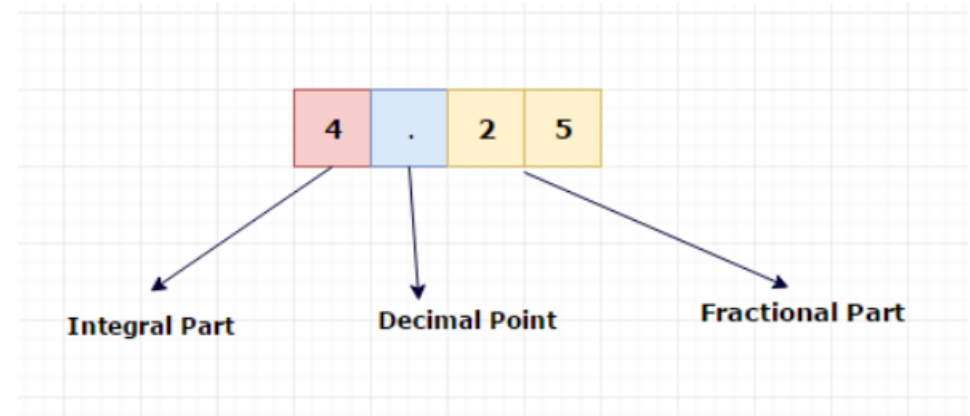
- 4 as $(100)_2$

- 0.25

$0.25 * 2 = 0.50$ //take 0 and move 0.50 to next step

$0.50 * 2 = 1.00$ //take 1 and stop the process

$0.25 = (01)_2$



Task

10.75?

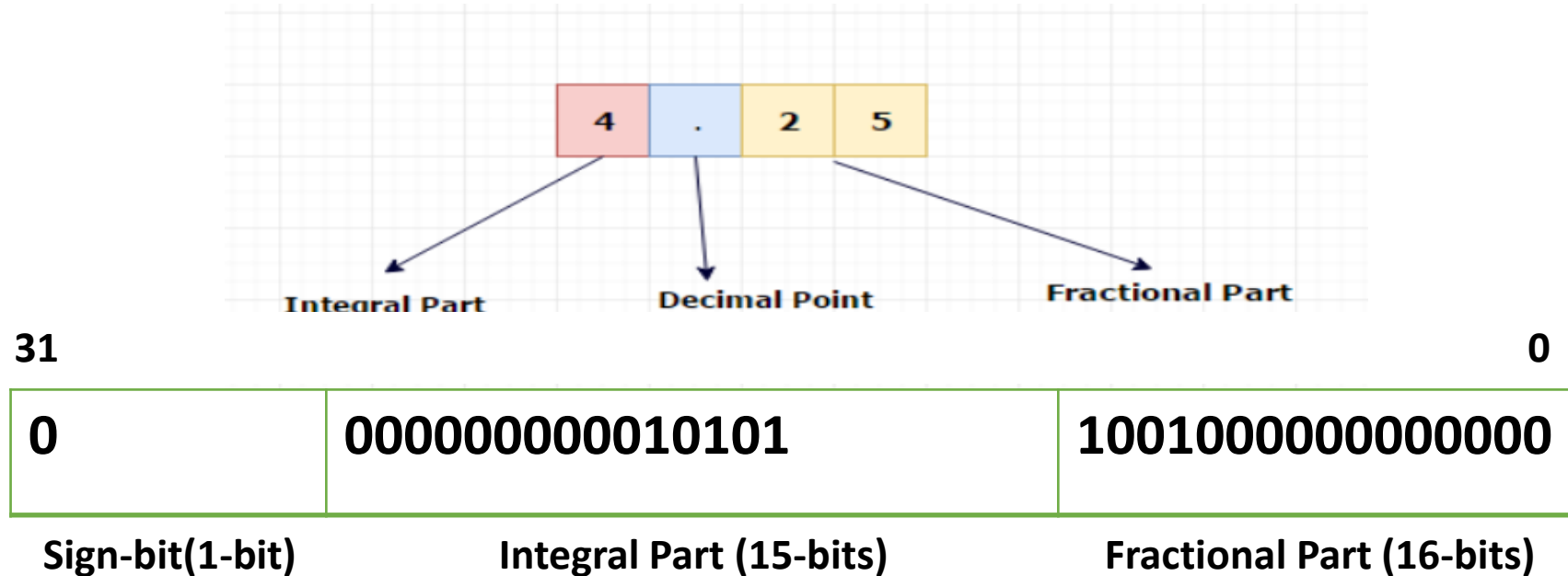
Fixed and Floating-Point Representation

Real Numbers

- Two major approaches to store real numbers
 - Fixed-Point Notation
 - Floating-Point Notation

Fixed-Point Representation

- Fixed number of bits for integer part and for fractional part.



Convert -43.625 to fixed number representation

1	000000000101011	1010000000000000
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Sign
bit

Integer part

Fractional part

Floating-Point Representation

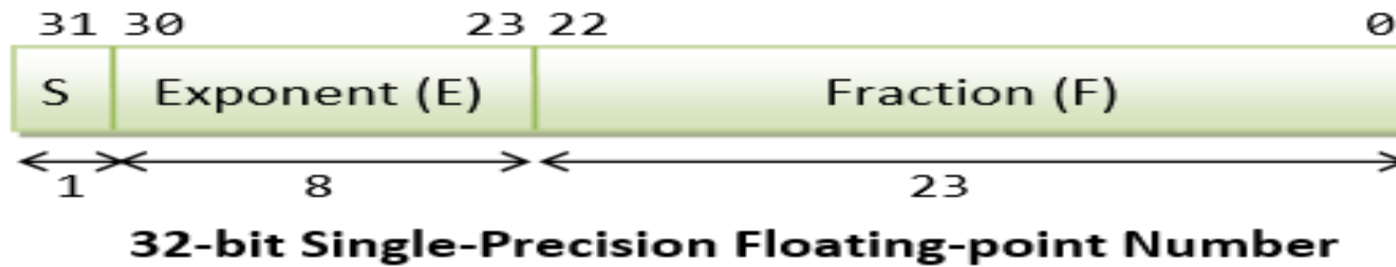
- Represent a number in the following form: $m \times r^e$.
 - Where m: mantissa
 - e: exponent
 - r: base/ radix
- Floating-point number is said to be normalized if the most significant digit of the mantissa is 1
 - $(-1)^s(1+m) \times 2^{(e-\text{Bias})}$



IEEE Floating point Number Representation

- Half Precision (16 bit): 1 sign bit, 5 bit exponent, and 10-bit mantissa
- Single Precision (32 bit): 1 sign bit, 8 bit exponent, and 23-bit mantissa
- Double Precision (64 bit): 1 sign bit, 11 bit exponent, and 52-bit mantissa
- Quadruple Precision (128 bit): 1 sign bit, 15 bit exponent, and 112-bit mantissa

32-bit Single-Precision Floating-Point Representation



32-bit Single-Precision Floating-Point Number

- Step 1: Convert decimal number to binary number
- Step 2: Normalize the number $(1.N)_2 \times 2^{E-127}$
- Step 3: Single-Precision Format

Example

- **Convert 299792458 to 32-bit single precision floating-point number**
- Step 1: 1000 1110 1111 0011 1100 0010 0101 0
- Step 2: $(1.000\ 1110\ 1111\ 0011\ 1100\ 0010\ 0101\ 0)^{155}$

$$E-127 = 28, E = 127+28 = 155$$

- Step 3:

0	10011011	000 1110 1111 0011 1100 0010 0101 0
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Task

- **Convert 3.1416 to 32-bit single precision floating-point number**

Arithmetic Operations

Binary Addition

- To perform the binary addition, we have to follow the binary table given below.
 - $0 + 0 = 0$
 - $0 + 1 = 1$
 - $1 + 0 = 1$
 - $1 + 1 = 0$ (carry 1 to the next significant bit)

E.g.:

$$\begin{array}{r} 111 \\ 110 \\ \hline 1001 \end{array}$$

$$\begin{array}{r} 1010 \\ 1101 \\ \hline 10111 \end{array}$$

$$\begin{array}{r} 11.01 \\ 101.11 \\ \hline 1001.00 \end{array}$$

- Add the binary numbers 00111 and 10101 and show the equivalent decimal addition.

$$\begin{array}{r} \textcolor{red}{0} \textcolor{red}{1} \textcolor{red}{1} \textcolor{red}{1} \\ 00111 \quad 7 \\ 10101 \quad 21 \\ \hline 11100 \textcolor{red}{=} 28 \end{array}$$

Binary Subtraction

- To perform the binary subtraction the following binary subtraction table should be followed.
 - $0 - 0 = 0$
 - $0 - 1 = 1$, borrow 1 from the next more significant bit
 - $1 - 0 = 1$
 - $1 - 1 = 0$

Decimal	Binary
11	1011
<u>5-</u>	<u>0101-</u>
<u>6</u>	<u>0110</u>

Payback (arrow from the 1 in the 1011 row to the 0 in the 0101 row)
 Borrow (arrow from the 0 in the 0101 row to the 1 in the 0110 row)

- Subtract the binary number 00111 from 10101 and show the equivalent decimal subtraction.

$$\begin{array}{r}
 \textcolor{red}{1} \textcolor{red}{1} \textcolor{red}{1} \\
 \textcolor{red}{1}0\textcolor{red}{1}01 \quad 21 \\
 \underline{00111} \quad \underline{7} \\
 01110 = 14
 \end{array}$$

1's Complement Subtraction

- Instead of subtracting the second number from the first, the 1's complement of the second number is added to the first number.
- The last carry which is said to be an END AROUND CARRY, is added to get the final result.
- If there is no end around carry find out the 1's complement of the sum and put a negative sign before the result as the result is negative

Example

- Subtract $(10000)_2$ from $(11010)_2$ using 1's complement.

Solution:-

$$\begin{array}{rcll} & 11010 & & = 26 \\ - & 10000 & \Rightarrow & \\ \hline & & + & \underline{01111} \text{ (1's complement)} & = -16 \\ \hline \text{Carry} \rightarrow & 101001 & & + 10 \\ + & \underline{1} & & \\ \hline & \underline{01010} & = & +10 \end{array}$$

Result is +10

2's Complement Subtraction

- In 2's complement subtraction, add the 2's complement of subtrahend to the minuend.
- If there is a carry out, ignore it.
- If the MSB is 0, the result is positive.
- If the MSB is 1, the result is negative and is in its 2's complement form. Then take its 2's complement to get the magnitude in binary

Example

For example:-

Subtract $(1010100)_2$ from $(1010100)_2$ using 2's complement.

Solution:-

$$\begin{array}{rcll} 1010100 & & 1010100 & = 84 \\ - 1010100 & \Rightarrow & + \underline{0101100} \text{ (2's complement)} & = - \underline{84} \\ & & 10000000 \text{ (Ignore the carry)} & \underline{0} \\ & = & 0 \text{ (result = 0)} & \end{array}$$

Hence MSB is 0. The answer is positive. So it is $+0000000 = 0$

Binary Multiplication

- The table for binary multiplication is given below
 - $0 \times 0 = 0$
 - $0 \times 1 = 0$
 - $1 \times 0 = 0$
 - $1 \times 1 = 1$

Example 1

1011 x 110

$$\begin{array}{r} 1011 \times \\ \underline{110} \\ 0000 \\ 1011 \\ \underline{1011} \\ \underline{1000010} \end{array}$$

Binary Division

- The table for binary division is as follows.
 - $0 \div 1 = 0$
 - $1 \div 1 = 1$

1) $1100 \div 11$

$$\begin{array}{r} 100 \\ 11 \overline{) 1100} \\ \underline{11} \\ 0 \end{array}$$

2) $1001 \div 10$

$$\begin{array}{r} 100.1 \\ 10 \overline{) 1001} \\ \underline{10} \\ 0010 \\ \underline{10} \\ 0 \end{array}$$

Divide $(10110)_2$ by $(110)_2$.

Solution :-

$$\begin{array}{r} 110 \) \ 101101 \ (\ 111.1 \\ - \ \underline{110} \\ 1010 \\ \underline{110} \\ 1001 \\ \underline{110} \\ 110 \\ \underline{110} \\ 000 \end{array}$$

Result is $(111.1)_2$

Complements in Number System

1. **1's Complement:** obtained by subtracting all bits from 1
2. **2's Complement:** 1's complement + 1
3. **7's Complement:** obtained by subtracting all bits from 7
4. **8's Complement:** 7's complement + 1
5. **9's Complement:** obtained by subtracting all bits from 9
6. **10's Complement:** 9's complement + 1
7. **15's Complement:** obtained by subtracting all bits from 15
8. **16's Complement:** 15's complement + 1

1's Complement Subtraction

Technique

- At first, find 1's complement of the B(subtrahend).
- Then add it to the A(minuend).
- If the final carry over of the sum is 1, then it is dropped and 1 is added to the result.
- If there is no carry over, then 1's complement of the sum is the final result, and it is negative.

1's Complement Subtraction

- **Find Subtraction of 110 and 101 using 1's complement method**
- **Find Subtraction of 10110 and 11101 using 1's complement method**

2's Complement Subtraction

- In 2's complement subtraction, add the 2's complement of subtrahend to the minuend.
- If there is a carry out, ignore it.
- If the MSB is 0, the result is positive.
- If the MSB is 1, the result is negative and is in its 2's complement form. Then take its 2's complement to get the magnitude in binary

- **Find Subtraction of 110 and 101 using 2's complement method**
- **Find Subtraction of 10110 and 11101 using 2's complement method**

7's Complement Subtraction

1. At first, find 7's complement of the B(subtrahend).
2. Then add it to the A(minuend).
3. If the final carry over of the sum is 1, then it is dropped and 1 is added to the result.
4. If there is no carry over, then 1's complement of the sum is the final result, and it is negative.

7's Complement Subtraction

- **Find Subtraction of 402 and 314 using 7's complement method**

8's Complement Subtraction

1. At first, find 8's complement of the B(subtrahend).
2. Then add it to the A(minuend).
3. If the final carry over of the sum is 1, then it is dropped, and the result is positive.
4. If there is no carry over, then 8's complement of the sum is the final result and it is negative.

8's Complement Subtraction

- **Find Subtraction of 342 and 614 using 8's complement method**

Any Questions

